Well, that's almost correct. Planck's law tells us that the energy of a single photon is proportional to its frequency (inversely proportional to its wavelength). By using light of lower and lower frequency (larger and larger wavelength) we can therefore produce ever gentler individual photons. But here's the catch. When we bounce a wave off of an object, the information we receive is only enough to determine the object's position to within a margin of error equal to the wave's wavelength. To get an intuitive feel for this important fact, imagine trying to pinpoint the location of a large, slightly submerged rock by the way it affects passing ocean waves. As the waves approach the rock, they form a nice orderly train of one up-and-down wave cycle followed by another. After passing by the rock, the individual wave cycles are distorted—the telltale sign of the submerged rock's presence. But like the finest set of tick marks on a ruler, the individual up-and-down wave cycles are the finest units making up the wave-train, and therefore by examining solely how they are disrupted we can determine the rock's location only to within a margin of error equal to the length of the wave cycles, that is, the wave's wavelength. In the case of light, the constituent photons are, roughly speaking, the individual wave cycles (with the height of the wave cycles being determined by the number of photons); a photon, therefore, can be used to pinpoint an object's location only to within a precision of one wavelength.

And so we are faced with a quantum-mechanical balancing act. If we use high-frequency (short wavelength) light we can locate an electron with greater precision. But high-frequency photons are very energetic and therefore sharply disturb the electron's velocity. If we use low-frequency (long wavelength) light we minimize the impact on the electron's motion, since the constituent photons have comparatively low energy, but we sacrifice precision in determining the electron's position. Heisenberg quantified this competition and found a mathematical relationship between the precision with which one measures the electron's position and the precision with which one measures its velocity. He found—in line with our discussion—that each is inversely proportional to the other: Greater precision in a position measurement necessarily entails greater imprecision in a velocity measurement, and vice versa. And of utmost importance,